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DRIVE SYSTEM INSTRUCTIONS

RIG OPERATION MANUAL

GENERAL ELECTRIC



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RIG OPERATION MANUAL
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THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN, ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

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WARNING Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

NOTE Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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INTRODUCTION

The purpose of this instruction is to guide operating personal on procedures to accomplish the following:

1. Operate the main switchboard controls
2. Make SCR/motor assignments
3. Control drilling motors
4. Control anchor windlass motors
5. Control propulsion motors
6. Operate the semi-submersible drilling rig

Figure 1 illustrates the physical pieces in the system and Figures 1A, 1B and 1C illustrate the system one-line for the main switchboard, drilling, anchor windlass, and propulsion system. Refer to these drawings often when reading system descriptions in the remainder of this guide.

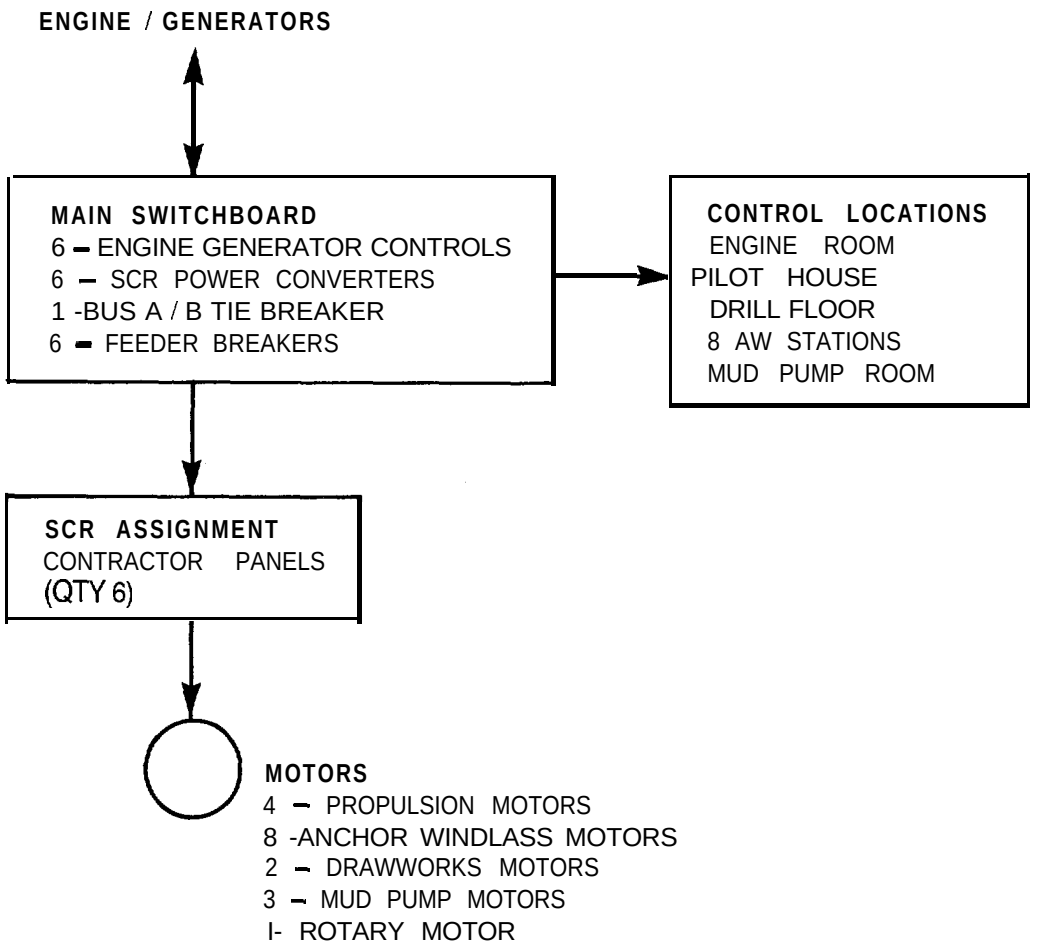


Figure 1. System Outline

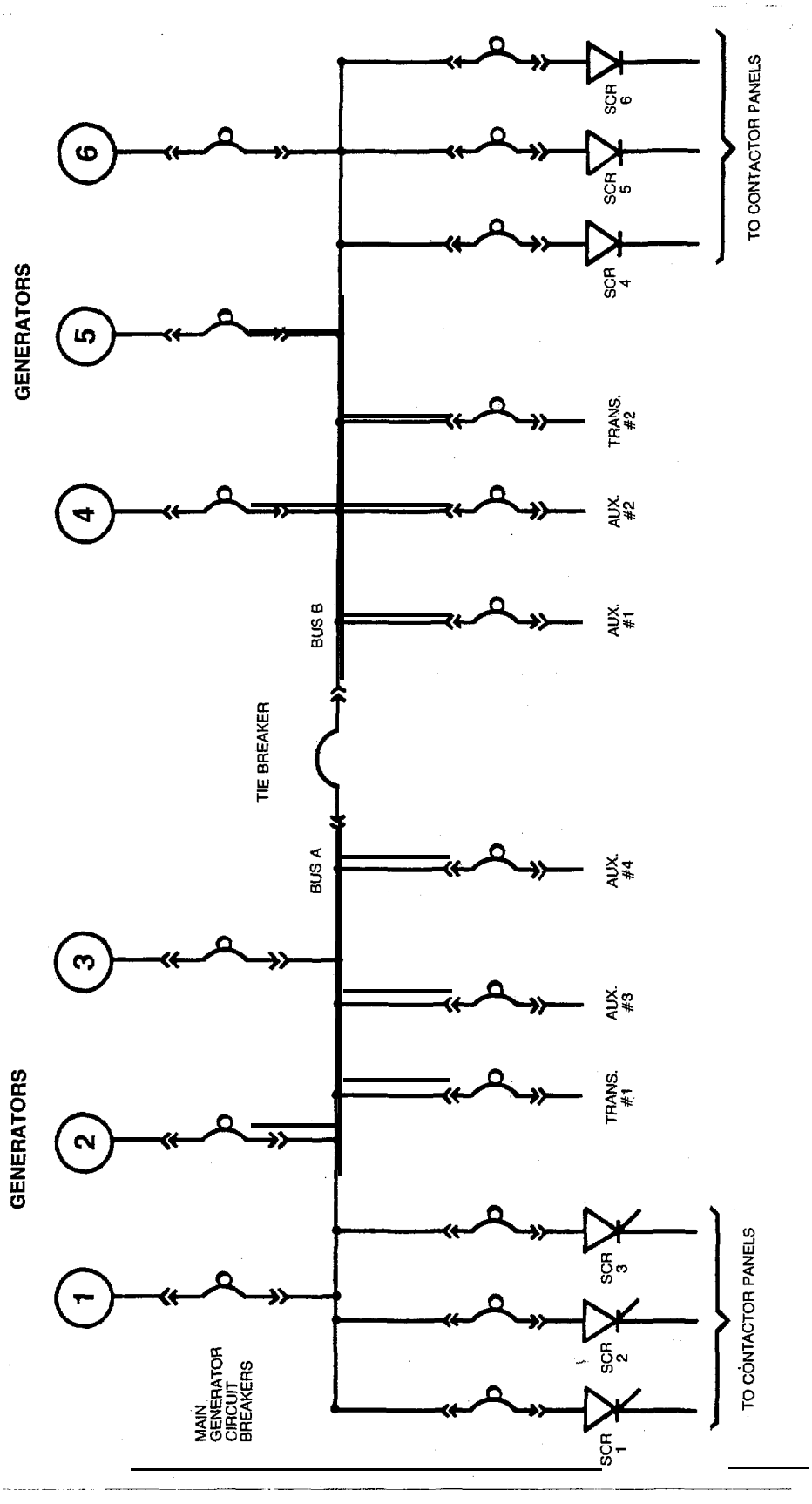


Figure 1A. Main Switchboard Line-up

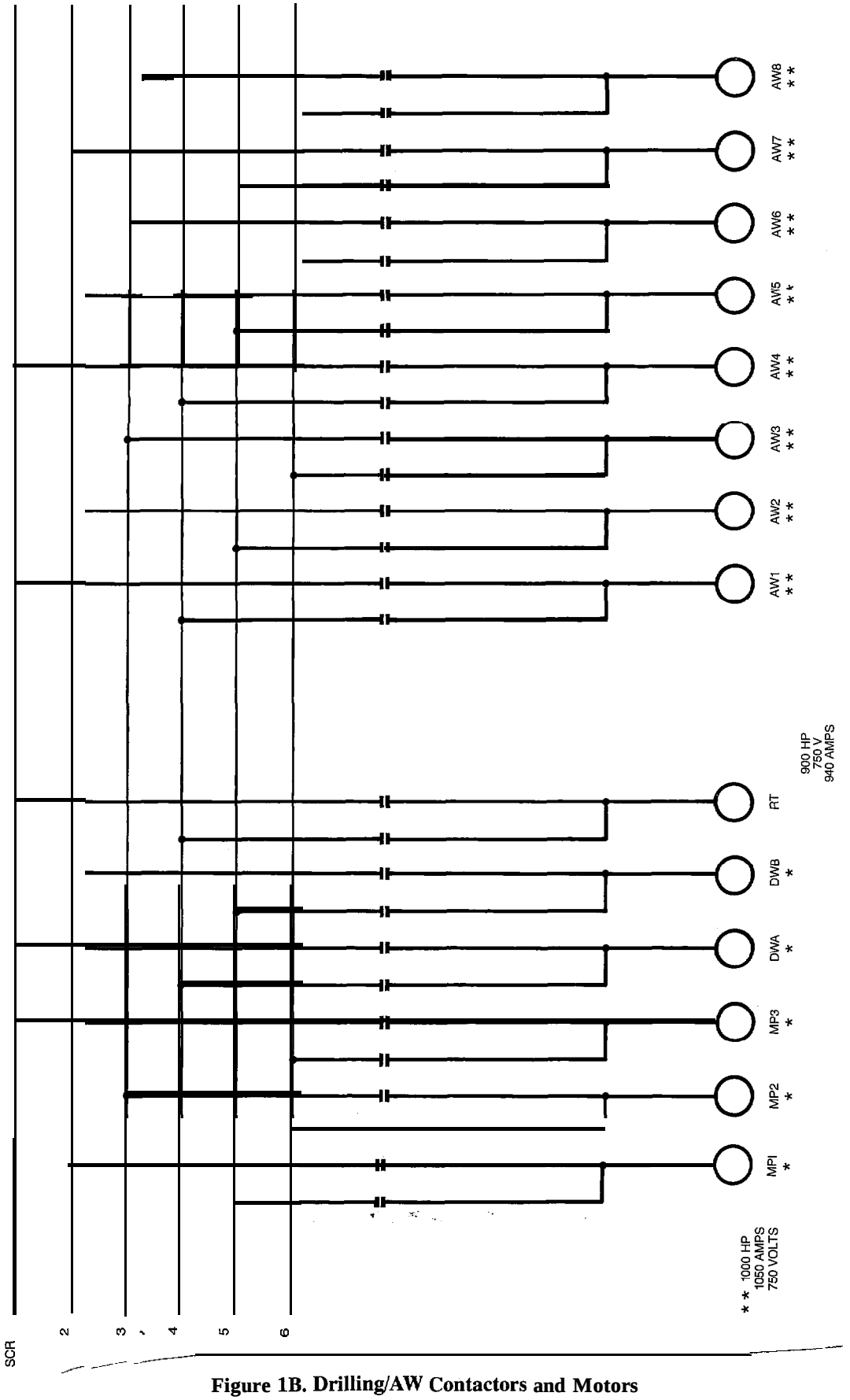


Figure 1B. Drilling/AW Contactors and Motors

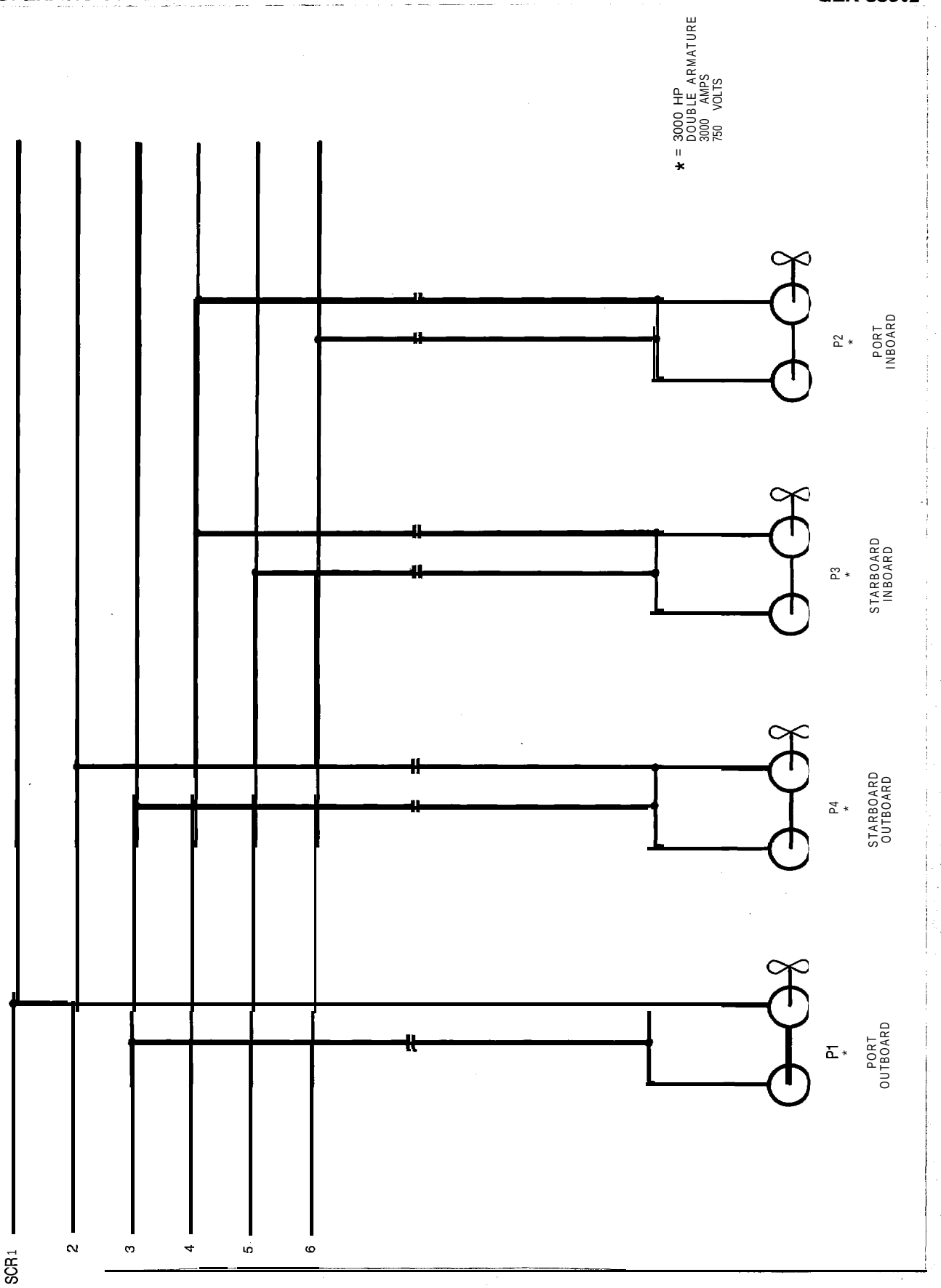


Figure 1C. Propulsion Contactors and Motors

CHAPTER 1
MAIN SWITCHBOARD

1-1. Control Equipment

The Main Switchboard equipment controls the engine/generator, converts three-phase AC power to DC power for the drilling motors, synchronizes the power supplied by several generators, and detects ground fault conditions in the AC or DC power circuits.

Most of the controls on the line-up are used during rig start up and shut down. The following panels and modules contain controls and indicators that the operator should be familiar with. Each of these panels and modules is described in this manual:

- Engine/Generator Control Panel
- Synchronizing Control
- Ground Detector
- Power Conversion Panel
- Feeder Breaker

1-1.1. Engine/Generator Control Panel

The Engine/Generator Control Panel consists of indicators, meters, and switches used to start and control a generator and its prime mover. There is one Engine/Generator Control Panel for each generator. The following devices are included on each panel:

1-1.1.1. Voltmeter. Indicates the AC voltage of a source selected by the voltmeter switch. The voltmeter can measure AC bus voltage or phase to phase voltage.

1-1.1.2. Ammeter. Indicates the current of the phase selected by the ammeter switch.

1-1 .1.3. Kilowatt Meter. Indicates engine loading.

1-1 .1.4. Kilovar Meter. Indicates generator loading.

1-1 .1.5. Engine Control Switch. Sets the operating mode of the engine/generator engine. In the OFF position, the engine is shut down. In the IDLE position, the engine operates at normal speed. In the LOAD position, the generator will load and the load sharing circuits of the generator are connected with the load sharing circuits of other on-line generators so that all generators will equally share the electrical load.

1-1 .1.6. Voltmeter Switch. Selects the voltage source to be displayed by the voltmeter. Selecting VL1-2 causes the voltmeter to display the voltage between phases 1 and 2 of the generator output. Likewise, selecting VL23 causes the voltmeter to display the voltage between phases 2 and 3 and selecting VL1 causes the voltmeter to display the voltage between phases 3 and 1. Selecting VBUSS causes the VOLTMETER to display the voltage of the AC bus (phases 3-l).

1-1 .1.7. Ammeter Switch. Selects the phase current to be displayed by the ammeter.

1-1 .1.8. Frequency Trim Knob. Adjusts engine speed to control the generator voltage frequency.

** NOTE **

The following engine/generator control lights indicate whether the generator is operating normally or whether a fault has occurred. The first fault will cause its associated light to flash. Any subsequent fault will cause its associated light to go out.

1-1.1.9. Gen On Light. When ON, indicates that the engine/generator is running, the breaker is closed, and power is being supplied to the AC bus. When OFF, indicates that the breaker is open.

1-1.1.10. Undervolt Light. Not used for this rig.

1-1 .1.11. No OV/UF Light. When ON, indicates a normal condition. When OFF, indicates that a generator overvoltage or underfrequency condition exists. The generator circuit breaker will trip for an overvoltage condition, but not for an underfrequency condition.

1-1 .1.12. No Rev Pwr Light. When ON, indicates a normal condition. When OFF, indicates that reverse power was flowing from the AC bus back to the generator.' This condition causes the generator circuit breaker to trip.

1-1.1.13. No Ovrspd Light. When ON, indicates a normal condition. When OFF, indicates that the generator engine was overspeeding. Overspeed causes fuel to be cut off from the engine and the generator breakers to open. ***The engine should be supplied with mechanical overspeed protection.***

1-1 .1.14. No Tach Loss Light , When ON, indicates a normal condition. When OFF, indicates that the tachometer signal has been lost. This condition causes

the generator engine to shut down and the generator breakers to open.

I-1 .I .15. Pwr Supply OK Light. When ON, indicates a normal condition. When OFF, indicates that the generator control power supply has malfunction. This condition causes the generator engine to shut down and the generator breakers to open.

I-I .1 .16. Generator Circuit Breaker. This circuit breaker connects the generator to the AC thru-bus. The CB closed from the synchronizing panel.

I-I .1.17. Auto/Manual Switch. Selects automatic or manual generator voltage regulation."Automatic" will normally be selected.

I-I .1.18. Manual Field Control. Sets generator voltage when Manual control is selected.

I-1.1.19. Space Heater Switch. Used to turn generator space heater on and off.

I-I .1.20. Space Heater "On" Light. Indicating light senses space heater voltage

I-I .1.21. RTD Meter. Meter indicates temperature of generator winding.

I-1.1.22. RTD Switch. Selects resistance temperature detectors in phase 1, 2, or 3 on generator in phase 1, 2, or 3 on generator. Can also be used to select "TEST" resistors.

1-1.2. Battery Module

This module is located in the generator control section of the line-up. It is used to energize the controls of the first engine/generator module to be started up. The battery module switch should always be in the ON position during rig operation so that the battery can be recharged. The switch must be turned OFF when the rig is shut down.

I-I .3. Synchronizing Control

The controls and indicators of the synchronizing control are located in the generator section of the line-up. The synchronizing control is used to synchronize the output frequency and phase of generators being brought on line with the frequency and phase of the 600 VAC bus. Failure to properly synchronize generators as they are brought on line can cause damage to the equipment. Two sets of controls are provided (BUS A / BUS B).

I-I .3.1. Frequency Switch. Selects the generator whose frequency is to be measured by the frequency meter.

I-1.3.2. Frequency Meter. Indicates the output frequency in Hertz of the generator selected by the frequency switch.

I-7.3.3. Synchronizing Switch. Selects the generator whose frequency is to be monitored by the synchroscope and connects the breaker closing circuit to the breaker close pushbutton.

I-I .3.4. Synchroscope. Indicates whether the output frequency of the generator selected by the synchronizing switch is faster or slower than the output frequency of the bus. Counterclockwise rotation of the needle means that the selected generator's output frequency is slower than the frequency of the bus. Clockwise rotation of the needle means that the selected generator's output frequency is faster than the frequency of the bus. A slow clockwise rotation is recommended for synchronizing. A separate synchroscope is provided for synchronizing the tie breaker.

I-1.3.5. Synchronizing Lights. Indicates the phase relationship between the generator selected by the synchronizing switch and the generator(s) already on line. The relative brightness of the lights indicates the degree that the generators are out of phase. When both lights are brightly lighted, the two generators are 180 degrees out of phase. When both lights are out, the two generators are in phase. Separate lights are provided for synchronizing tie breaker.

I-1.3.6. Breaker Close Button. When pressed, closes the circuit breaker of the generator selected by the synchronizing switch and brings the generator output on line to the 600 VAC bus. The generator's circuit breaker must be in the CHARGED position before the breaker close pushbutton will work.

I-1.3.7. Bus Frequency Meter. Frequency meter dedicated to the AC thru bus.

I-I .3.8. Bus Voltmeter. Voltmeter dedicated to the AC thru bus.

I-1.3.9. Tie Breaker Sync Switch. Selects tie breaker synchroscope and lights.

I-1.3.10. Tie CB. The tie circuit breaker connects Bus A to Bus B. A synchroscope and lights are provided for manually synchronizing this circuit breaker.

I-1.3.1 1. 24 VAC Control Power Switch. This switch must be set to an energized bus (A or B) to transform 600 VAC ti 24 VAC for the operator consoles.

I-1.4. Ground Detector Module

This module detects ground faults within the AC and DC power system and displays the faults on the lights described below. A lamp test button allows the operator to check the lamps. A contact is provided for remote alarm. One ground detector is provided on each bus.

I-1.4.1. AC Ground Detector. These three lights correspond to the three phases present on the 600 volt AC bus. The lights are normally on: when a phase is grounded, the corresponding light turns off.

I-I .4.2. DC Ground Detector. These two lights correspond to the positive and negative SCR DC buses. The lights are normally off: when a bus is grounded, the corresponding light comes on.

1-1 .4.3. Lamp Test Button. Pressing this button causes all AC ground detector lamps to light. This should be done periodically to detect faulty lamps.

I-I .5. Power Conversion Panel

The power conversion panel uses SCRs (Silicon Controlled Rectifiers) to convert AC voltage supplied by the generators to a variable DC voltage for the drilling motors. The front of this panel contains two indicators, two meters, and a reset button that monitor the operation of the SCRs.

1-1 .5.1. Voltmeter. This *meter* indicates the DC voltage supplied to the drilling motor by the SCR.

1-1 .5.2. Ammeter. This meter indicates the DC current supplied to the drilling motor by the SCR.

1-1 .5.3. SCR OK Light. When On, indicates that the SCR is available for drive selection.

1-1.5.4. Drive Ready Light. When on, indicates that the drive is ready to run. When off, indicates that the drive is not selected. When flashing, indicates that the selected drive is awaiting permissives or indicates a warning if the drive is running.

1-1 .5.5. Reset Pushbutton. Pressing *this* button causes the power conversion panel to check itself for shorted cells or failure of a cell to fire. This button is active only when the drive is not selected.

I-1.6. Feeders

Feeder breakers are provided to distribute 600 VAC power to customer supplied equipment.

CHAPTER 2

2-1. SCR To Motor Assignments

There are three main control locations on the ODECO drilling platform. These are Engine Room, Pilot House, and Driller's console. Anchor windlasses will each be controlled from their individual control plates. Also, there is a mud pump console plate located near the pumps. The console layouts are included in Chapters 3, 4, and 5. Refer to these layouts to find the switches and lights mentioned below.

In normal operating mode, the Driller's console and the Engine Room control have equal access to the SCRs in the system and may each make assignments as needed. This sharing of assignment control is governed by lights on each console which indicate which SCRs are available for assignment.

For example, initially no SCRs will be assigned. The driller may want to use the drawworks. He will look at his console and find that all the "SCR Available" lights are lighted. This tells him that he may make any valid SCR assignment to the drawworks motor. Let us say he assigns SCR No. 1 to Drawworks Motor A. Upon making the assignment, the "SCR No. 1 Available" lights in the Driller's console and on the engine room console will go out and the "Drawworks Motor A Ready" light on the Driller's console will come on. This indicates that the motor will now respond to the Drawworks throttle and all permissives to run are satisfied. SCR No. 1 is now occupied by an assignment. The operator in the Engine room knows that since the "SCR No. 1 Available" light is off, he cannot assign an Anchor windlass or a propulsion motor to SCR No. 1.

The above scheme also works for assignments made in the Engine room console. If SCR No. 2 were assigned in the Engine room to Anchor Windlass No. 2, the "SCR No. 2 Available" in both driller's and engine room consoles would go out. The driller would then know that he could not assign a motor to SCR No. 2. The "Anchor Windlass No. 2 Ready" light would light at the Anchor Windlass No. 2 console to indicate that the windlass is ready for operation.

This plan of operation gives the advantage of being able to run drilling and propulsion functions simultaneously. This would allow maintenance of minor positioning functions to be performed without interrupting the drilling operation.

The master control location for SCR assignment and control is in the Engine room. Therefore, the Engine Room operator can turn an override switch and cause the Engine Room control to take priority over all other rig functions. In this mode, only the Engine Room can make SCR assignments.

The SCR/Motor assignment chart is shown in Table 1.

Table 1. Thyristor Drilling Control

		SCR Motor Assignments					
		SCR1	SCR2	SCR3	SCR4	SCR5	SCR6
	Motors						
	MP1		x			x	
	Mud Pump						
	MP2			x			x
	MP3	x					x
	Drawworks						
	DWA	x			x		
	DWB		x			x	
	Rotary Table						
	RT	x			x		
	Anchor Windlass						
	AW1	x			x		
	AW2		x			x	
	AW3			x			x
	AW4	x			x		
	AW5		x			x	
	AW6			x			x
	AW7		x			x	
	AW8			x			x
	Propulsion						
	P1	x		x			
(4)	P2				x		x
	P3				x	x	
(4)	P4		x	x			

CHAPTER 3
CONTROL OF DRILLING MOTORS

3-4. ASSIGNMENT

Once the main switchboard and SCR Powered Conversion Panels are operational the drilling motors can be assigned and controlled from the drillers console.

The mud pump motors may also be controlled from a remote MP location (see 3-3, 3-4)

3-2. DRILLER'S CONSOLE (See sh. 3-2)

The indicators, meters, throttles and switches on the driller's console give the driller control over the drilling motors. A description of each item on the driller's console is given below.

3-2.1. STATUS LIGHTS

These lights show the operating condition of rig equipment such as generators, SCRs, and motors. Each light is labelled with the name of the equipment or function monitored. When the light is on, the equipment or function is active. Some typical items that may be monitored by the status lights are listed below.

3-2.1.1. Gen On Device #17-22. The indicated generator is supplying power to the drive system when this light is on.

3-2.1.2. Drive Ready 5-7, 33-35. The drive is selected and ready to run when this light is on. When the light flashes, it indicates that the drive is selected and awaiting permissives or indicates a warning if the drive is running.

3-2.1.3. Supercharger On 11, 12. The supercharger is running when this light is on. When the light flashes, it indicates the supercharger failed to start.

3-2.1.4. M.P. Auxiliaries 8-10. The mudpump auxiliary motors are running when this light is on. When the light flashes, it indicates the auxiliary motors failed to start.

3-2.1.5. High Load 29. The engine/generator sets supplying power to the rig are approaching an overload condition when this light comes on.

3-2.1.6. Alarm Silenced 32. The alarm is silenced when this light is flashing. In this condition, the audible alarm will not sound for abnormal conditions.

3-2.1.7. SCR Available 23-28. When lit, the SCR is available for drilling motor assignment. This light will go out when SCR is assigned.

3-2.1.8. Ground Fault 30. Indicates AC or DC ground fault

3-2.1.9. Loss Purge 31. Illuminates when the drillers console loses purging air.

3-2.1.10. Prop Override 51. This light indicates that the Engine Room has taken control away from the drillers console.

3-2.1 .11. High/Low Clutch 53-54. Drawworks clutch position indication.

3-2.2. SELECTION SWITCHES 36-41

These switches allow the driller to select specific drilling motor functions and directions. There is one switch per SCR, where each switch can select at least one of the following functions.

3-2.2.1. Mud Pump. A mud pump is selected by turning the SCR selection switch to the MPX position. Control of the mud pump is transferred to the mud pump console by turning the switch to the REM position. (Device l-3)

3-2.2.2. Drawworks. The drawworks are enabled when the SCR selector switch is in the DWX position. The driller must also select FWD or REV. (Device 42)

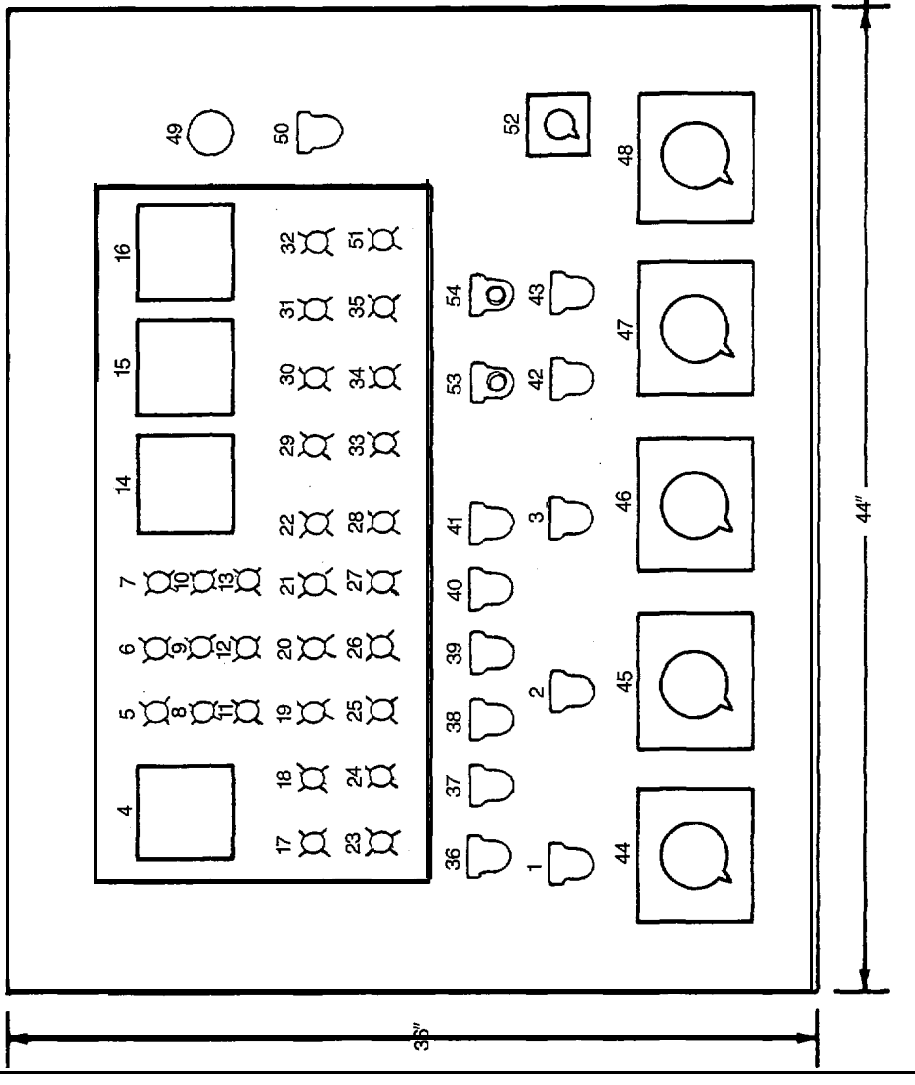
3-2.2.3. Rotary Table. The rotary table is enabled when the SCR sector switch is in RT position. The driller must also select FWD or REV. (Device 43)

3-2.3. THROTTLES (44-48)

The throttles control the speed of the motors indicated on the respective console nameplates. The throttles must be off to assign the drive.

3-2.4. ROTARY TABLE TORQUE LIMIT CONTROL (52)

This control limits the torque 'provided to the rotary table. The rotary table may be chain-driven by the drawworks motor or it may be an independent rotary with its own motor.



	NAMEPLATE	FUNCTION
1.	"MP1, Remote -- Off = On"	Selector switch
2.	"MP2. Remote = Off-On"	Selector switch
3.	"MP3. Remote = Off On"	Selector switch
4.	"% KW, 0-100%"	KW meter
5.	"MP1 Ready"	Grn ind light
6.	"MP2 Ready"	Grn ind light
7.	"MP3 Ready"	Grn ind light
8.	"MP1 Aux"	Grn ind light
9.	"MP2 Aux"	Grn ind light
10.	"MP3 Aux"	Grn ind light
11.	"MP1 S.C."	Grn ind light
12.	"MP2 S.C."	Grn ind light
13.	"Spare"	Grn ind light
14.	DWA, O-2500	Ammeter
15.	DWS, O-2500	Ammeter
16.	RT, O-1500	Ammeter
17.	"Gen #1 On"	Red ind light
18.	"Gen #2 On"	Red ind light
19.	"Gen #3 On"	Red ind light
20.	"Gen #4 On"	Red ind light
21.	"Gen #5 On"	Red ind light
22.	"Gen #6 On"	Red ind light
23.	"SCR #1 Avail"	Grn ind light
24.	"SCR #2 Avail"	Grn ind light
25.	"SCR #3 Avail"	Grn ind light
26.	"SCR #4 Avail"	Grn ind light
27.	"SCR #5 Avail"	Grn ind light
28.	"SCR #6 Avail"	Grn ind light
29.	"High Load"	Red ind light
30.	"Grd. Fault"	Grn ind light
31.	"Loss Purge"	Red ind light
32.	"Alarm Sil. On"	Red ind light
33.	"DWA Ready"	Grn ind light
34.	"DWB Ready"	Grn ind light
35.	"RT Ready"	Grn ind light
36.	"SCR1, MP3-Off-DWA-RT"	Selector switch
37.	"SCR2, MP1-Off-DWB"	Selector switch
38.	"SCR3, Oft-MPT	Selector switch
39.	"SCR4, DWA-Off-RT"	Selector switch
40.	"SCR5, MP1-Off-DWB"	Selector switch
41.	"SCR6, MP2-Off-MP3"	Selector switch
42.	"Drawworks" Rev-Off-Fwd	Selector switch
43.	"Rotary" Rev-Off-Fwd	Selector switch
44.	"MP1"	Throttle rheo
45.	"MP2"	Throttle rheo
46.	"MP3"	Throttle rheo
47.	"DW"	Throttle rheo
48.	"RT"	Throttle rheo
49.	"Emergency Shutdown"	Pushbutton
50.	"Alarm Silence"	Selector switch
51.	Silence-On	White ind light
52.	Prop Override	Rheostat
53.	R.T. Current Limit	White ind light
54.	High Clutch	White ind light
	Low Clutch	White ind light

Figure 2. Driller's Console

3-2.5. EMERGENCY SHUTDOWN (49)

Pressing this button trips the SCR disconnects, thus removing motor field and armature power. *Otherpower may still be connected to the motor at this time!.*

3-2.6. %KW METER (4)

This meter indicates how much of the total power available from the engines is being used by the rig. Using this meter, an operator can determine the correct number of engines to run to supply enough power for rig operation. The position of the needle indicates the number of engines required.

3-2.7. DRAWWORKS AND ROTARY TABLE AMMETERS (14-16)

This meter indicates the amount of amps flowing through the respective motor.

3-2.8. ALARM SILENCE SWITCH (50)

When activated, this switch prevents the audible alarm from sounding at the driller's console. The alarm silenced light will flash as long as this switch is ON.

3-2.9. AUDIBLE ALARM

Unless disabled by the alarm silence switch, the audible alarm sounds in case of:

1. Loss of a motor blower while the motor is running.
2. Loss of the supercharger while a mud pump motor is running.
3. Loss of a mud pump auxiliary motor (i.e., a lube motor) while the mud pump motor is running.

3-3. MUD PUMP CONSOLE (See Figure 3)

The mud pump console is simply an extension of the driller's console and is used to control the mud pumps from a remote location.

3-3.1. THROTTLES

The mud pump throttles serve the same purpose on the mud pump console as on the driller's console: controlling the speed of the mud pump motors. Device #20, 21, *22*

3-3.2. STATUS LIGHTS

3-3.2.1. MP Ready 3, 7, **11.** The following conditions must be met:

1. The remote throttle must be selected by turning the mud pump selection switch on the driller's console to the REMOTE position.
2. The auxiliary and supercharger motors must be operating.
3. The power converter must be operating.

3-3.2.2. MP Control Here Indicating Light 2, 6, 10 When this light is on, the drivers console MP control selector switch is in the "Remote" position and the mud pump console has control of the selected mud pump.

3-3.2.3. MP Auxiliary Indicating Light 4, 8, 12. Same function as drillers console auxiliary status light.

3-3.2.4. MP Super Charger Indicating Light 5, 9

Same function as drillers console MP supercharger. Status light

3-3.2.5. Loss Purge 1. Indicates when the mud pump console has lost purging air.

3-3.3. SELECTOR SWITCHES

3-3.3.1. Mud Pump Off-On Selector Switch 17, 18, 19. Selects/Deselects a mud pump for operation. Mud pump control must be available at the mud pump console

3-3.4. Meters

3-3.4.1. MP Ammeters. Indicate amperes flowing through each mud pump motor.

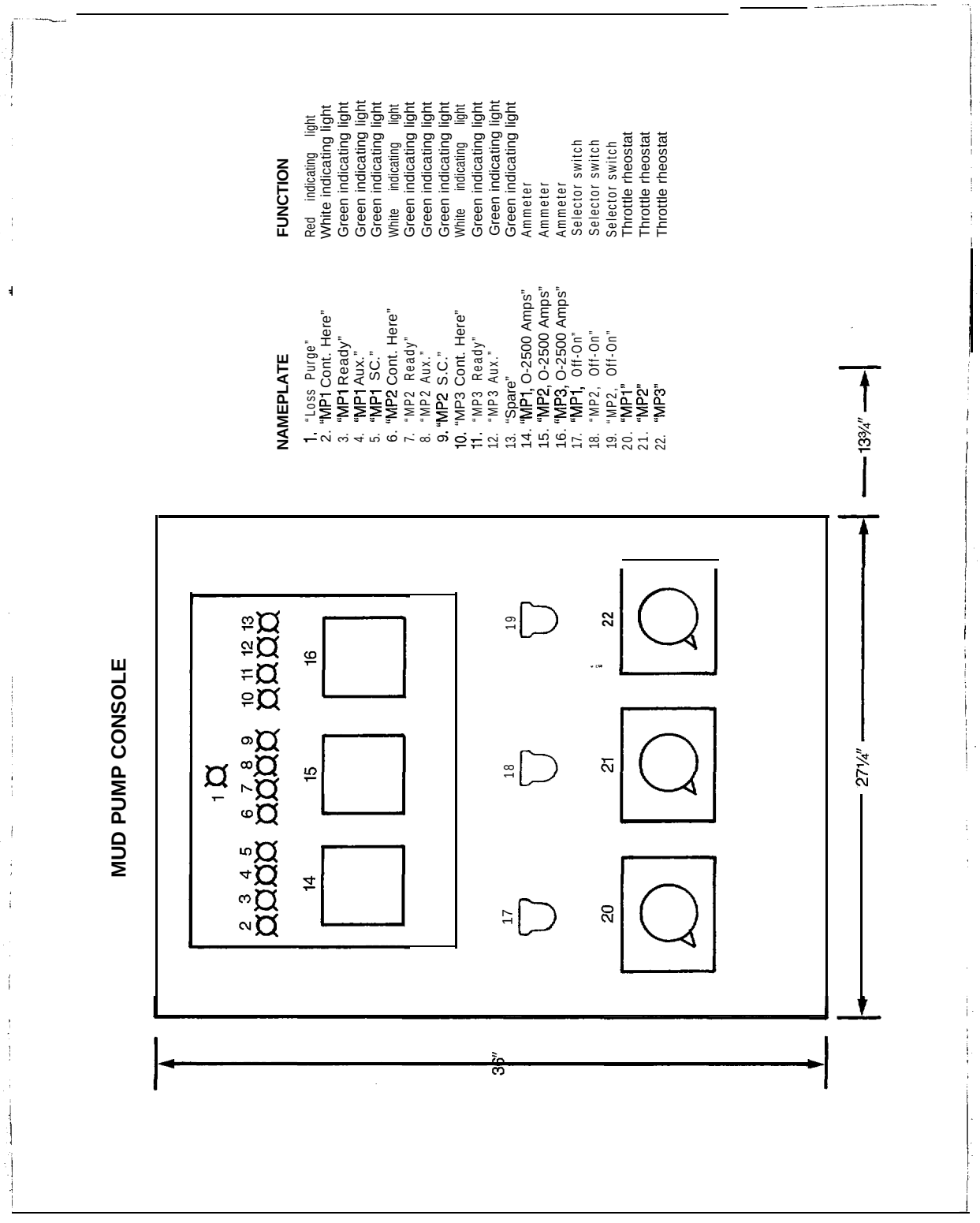


Figure 3. Mud Pump Console

CHAPTER 4
ANCHOR WINDLASS CONTROL

4-1. ASSIGNMENT

Each anchor windlass motor can be assigned to one of two SCR's. This assignment is made at the engine room console. Once assigned, the corresponding "SCR available" light will turn off at both the engine room console and at the driller's console, indicating that the SCR is in use.

Refer to the AW device plate layout, Figure 4, when reading the descriptions below.

4-2. ANCHOR WINDLASS OPERATOR'S DEVICE PLATE (See Figure 4)

4-2.1. STATUS LIGHTS

4-2.1.1. Ready Light Device #: 2. The drive is selected at the engine room console, and ready to run when this'light is on. When the light flashes, the drive is selected and waiting for permissives or indicates a warning if the drive is running.

4-2.2. Switches and Pushbuttons

4-2.2.1. Jog Pushbutton 3. When pushed, the motor brake will release and motor will run slowly in the haul-in direction.

4-2.2.2. Brake By-Pass Pushbutton 1. Pushbutton can be pressed to release the motor brake.

4-2.2.3. Wildcat-Off-Drum Selector Switch 4. The operator must select "Wildcat" or "Drum" before operating the anchor windlass. This switch determines proper motor direction. Refer to AW Manufacturer's Instructions.

4-2.2.4. Normal Payout Selector Switch 6. This switch selects **Normal** mode of operation or "Maintenance Payout" of the anchor windlass. The switch will spring return to normal.

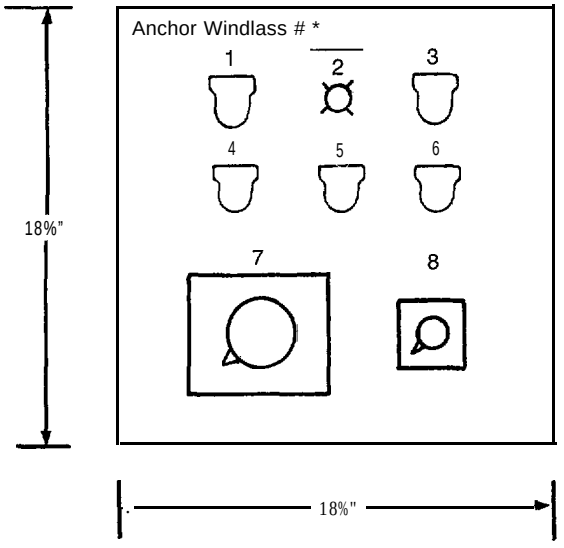
4-2.2.5. Haul-In/Off/Payout Selector Switch 5. Selects Haul-in, OFF, or regenerative payout mode during Normal operating.

4-2.3. RHEOSTATS

4-2.3.1. AW Throttle 7. The throttle controls the speed of the anchor windlass motor.

4-2.3.2. Current Limit 8. Motor current limit is adjustable by the operator from 0-100%.

ANCHOR WINDLASS OPERATOR'S PLATE



NAMEPLATE	DESCRIPTION
1. "Brake By-Pass"	Pushbutton
2. "Ready"	Green indicating light
3. "Motor Jog"	Pushbutton
4. "Wildcat-Off-Drum"	Selector switch
5. "Haul-In-Off-Payout"	Selector switch
6. "Normal-Payout"	Selector switch (Key lock switch. Key must be turned for payout. Spring return to Normal.)
7. AW *	Throttle rheostat
8. "Current-Limit"	Rheostat

• NOTE-Quantity eight (8) operators plates to be furnished and will be numbered 1 through 8.

Figure 4. Anchor Windlass Operator's Plate

CHAPTER 5
PROPULSION CONTROL

5-1. ASSIGNMENT .

Once the main switchboard and SCR power conversion panels have been powered up, the propulsion motors may be assigned at the engine room console and controlled from either the engine room or pilot house.

5-1.1. ENGINE ROOM OPERATORS DEVICE PLATE (See Figure 4 for layout.)

5-1.1.1. SCR Available Indicating Light Device #: 13 - 18. When ON, The SCR is available for assignment. This light will turn off when the SCR is assigned,

5-1.1.2. SCR Assignment Switch 19-24. One switch per SCR assigns either an anchor windlass or propulsion motor.

5-1.1.3. Here - Pilot House Switch 26. Selects engine room for propulsion control location or requests transfer to pilot house.

5-1.1.4. Control Here Indicating Light 25. Indicates propulsion control is active at Engine Room.

5-1.1.5. Control Pilot House Indicating Light 27 Indicates propulsion control is active at Pilot House

5-1 .1.6. Normal - Propulsion Override Selector Switch 28. When propulsion override is selected, drilling assignments are disabled at Drillers Console.

5-l .1.7. Propulsion Ready Indicating Light9-12 . Drive ready light for propulsion motors. Indicates that propulsion drive is assigned, permissives are met and drive is Ready to run. Light will flash if motor is selected and permissive is not met or if drive is running with a warning.

5-l .1.8. Propulsion Throttles 29-32. Control speed and direction of propulsion motor when Engine room is in control. Must be OFF to assign drive.

5-1.1.9. Propulsion Speed Meters 1, 3, 5, 7. Indicate propeller speed.

5-1.1.10. Propulsion Ammeters 2, 4, 6, 8. Indicate amperes flowing through propulsion motors.

5-1.2. Pilot House Operators Device Plate (See Figure 5 for layout)

5-1.2.1. Propulsion Ready Indicating Light 9, 10, 11,12. Same function as Engine Room "Propulsion Ready" Lights.

5-l .2.2. Propulsion Power Available Indicating Light 9A, 10A, 1 IA, 12A. Indicates that power is available for the propulsion motor.

5-1.2.3. Engine Room Control Indicating Light 17 On when Engine Room is in control.

5-1.2.4. Pilot House Control Indicating Light 18. On when Pilot House is in control.

5-1.2.5. Take Control Indicating Light 19. Light turns ON when Engine Room requests transfer of control from Engine Room to Pilot House with control location selector switch.

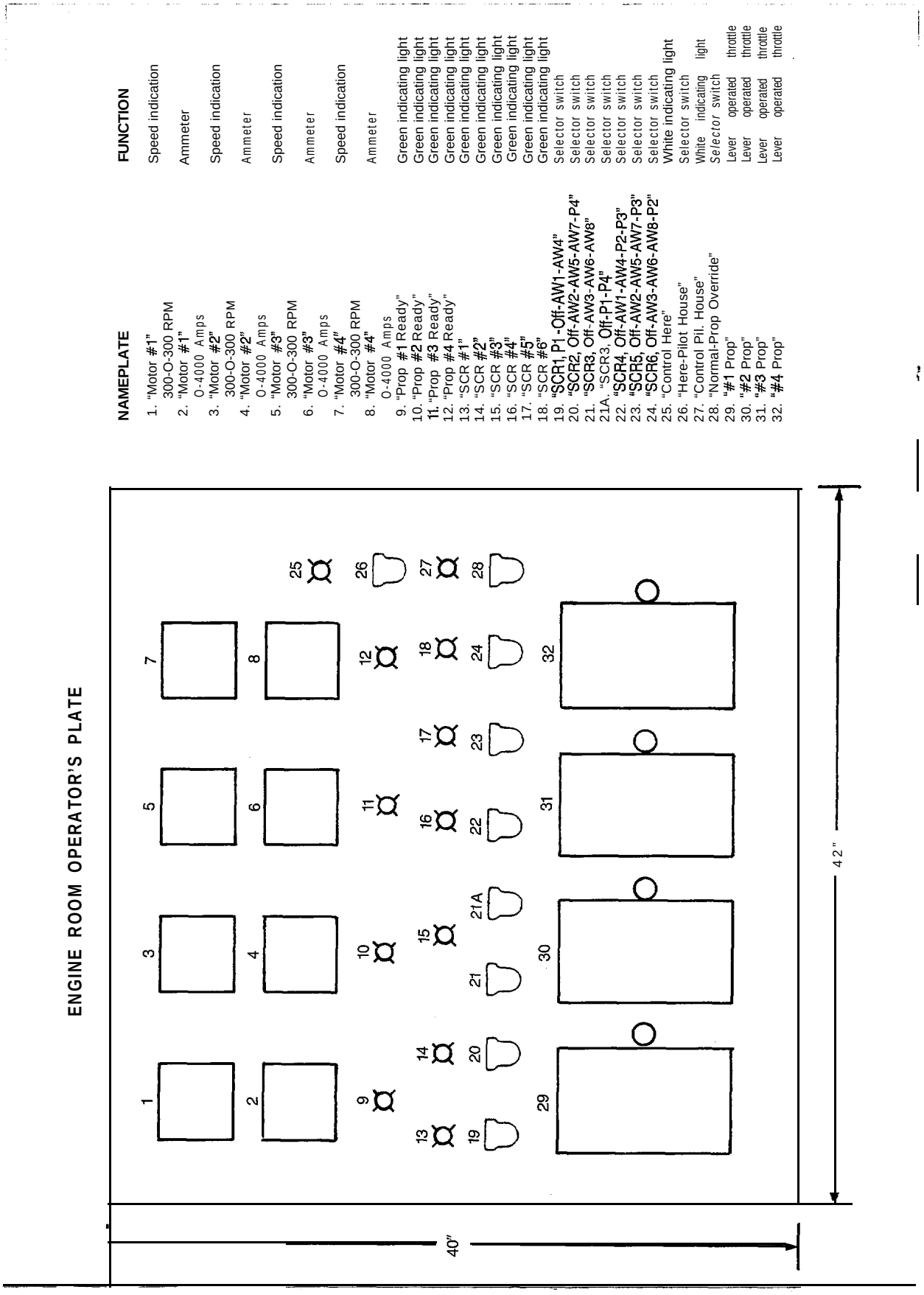
5-1.2.6. Take Control Pushbutton 20. PRESS to acknowledge Engine Room to Pilot House control transfer.

5-1.2.7. Dimmer 21. Light Dimmer control for indicating lights and illuminated meters.

5-1.2.8. Propulsion Throttle 13-1 6. Controls speed and direction of propulsion motors when pilot house is in control. Throttle must be OFF to assign drive.

5-1.2.9. Propulsion Speed Meters 1, 3, 5, 7. Indicate propeller speed.

5-1.2.10. Propulsion Ammeters 2, 4, 6, 8. Indicate amperes flowing through propulsion motors.



	NAMEPLATE	FUNCTION
1.	"Motor #1"	Speed indication
	300-O-300 RPM	
2.	"Motor #1"	Ammeter
	0-4000 Amps	
3.	"Motor #2"	Speed indication
	300-O-300 RPM	
4.	"Motor #2"	Ammeter
	0-4000 Amps	
5.	"Motor #3"	Speed indication
	300-O-300 RPM	
6.	"Motor #3"	Ammeter
	0-4000 Amps	
7.	"Motor #4"	Speed indication
	300-O-300 RPM	
8.	"Motor #4"	Ammeter
	0-4000 Amps	
9.	"Prop #1 Ready"	Green indicating light
10.	"Prop #2 Ready"	Green indicating light
11.	"Prop #3 Ready"	Green indicating light
12.	"Prop #4 Ready"	Green indicating light
13.	"SCR #1"	Green indicating light
14.	"SCR #2"	Green indicating light
15.	"SCR #3"	Green indicating light
16.	"SCR #4"	Green indicating light
17.	"SCR #5"	Green indicating light
18.	"SCR #6"	Green indicating light
19.	"SCR1, P1-Off-AW1-AW4"	Selector switch
20.	"SCR2, Off-AW2-AW5-AW7-P4"	Selector switch
21.	"SCR3, Off-AW3-AW6-AW8"	Selector switch
21A.	"SCR3, Off-P1-P4"	Selector switch
22.	"SCR4, Off-AW1-AW4-P2-P3"	Selector switch
23.	"SCR5, Off-AW2-AW5-AW7-P3"	Selector switch
24.	"SCR6, Off-AW3-AW6-AW8-P2"	Selector switch
25.	"Control Here"	White indicating light
26.	"Here-Pilot House"	Selector switch
27.	"Control Pil. House"	White indicating light
28.	"Normal-Prop Override"	Selector switch
29.	"#1 Prop"	Lever operated throttle
30.	"#2 Prop"	Lever operated throttle
31.	"#3 Prop"	Lever operated throttle
32.	"#4 Prop"	Lever operated throttle

Figure 5. Engine Room Operator's Plate

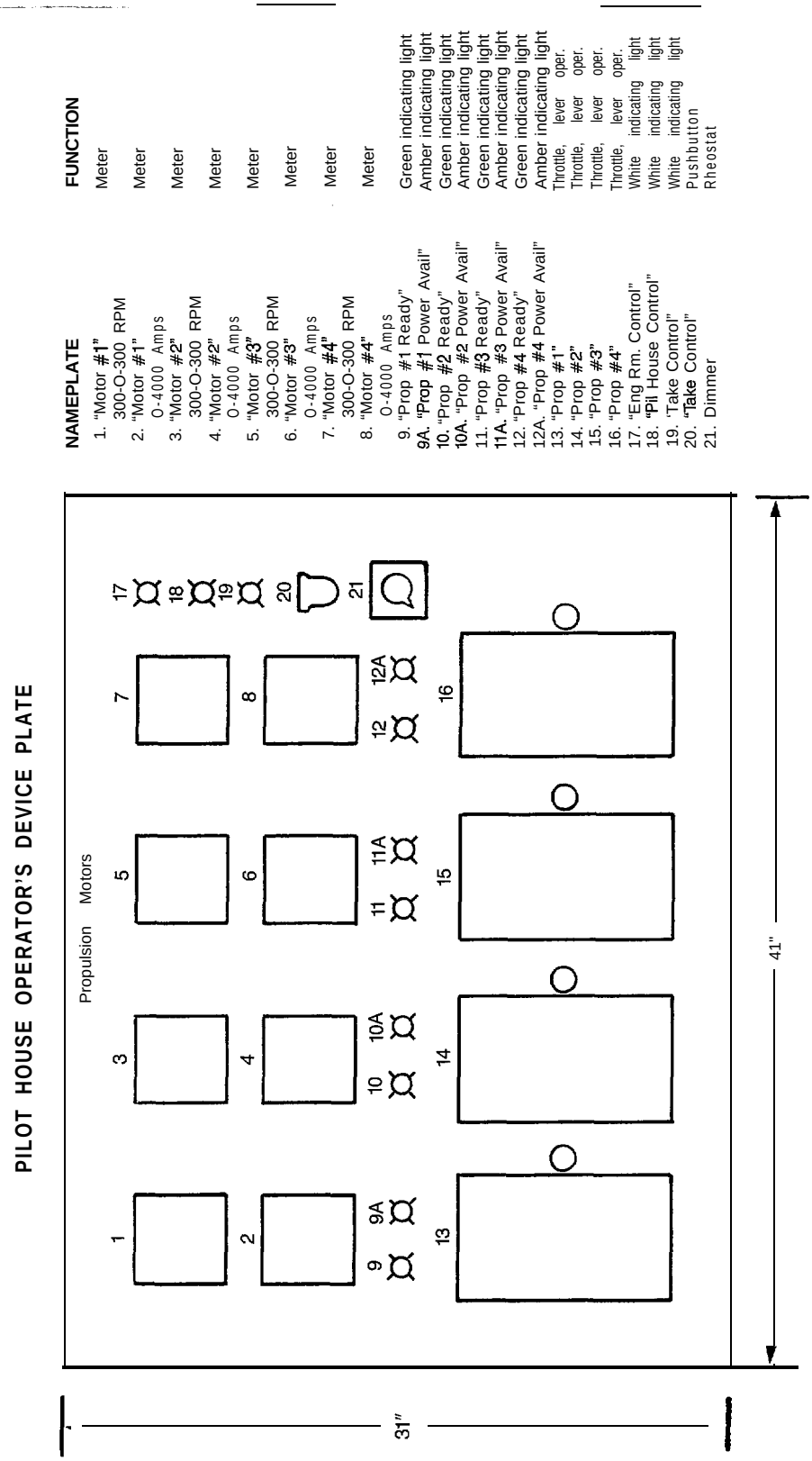


Figure 6. Pilot House Operator's Device Plate

CHAPTER 6
OPERATING PROCEDURES

The procedures outlined below are for normal, day-by-day operation of the rig drive system, including rig start up, operation, and shut down.

6-1. RIG START UP

This procedure describes how to start up a rig drive system, assuming that the rig, drive system, generators, and engines have been properly installed and checked out.

1. Preliminary steps:
   a. Ensure that the emergency stop switches on the driller's console are pulled out.
   b. Deselect all motors and turn all throttles off at all control locations.
   c. Position the battery module power switch to the ON position.
   d. Ensure that the engine control switch for the generator being started is in the IDLE position.
2. Start the desired engine and follow the manufacturer's recommended warm up procedures before continuing with the following steps. (The following steps assume that generator 1 is being brought on line first, followed by generator 2.)
3. Set the synchronizing switch to GEN 1.
4. Set the generator 1 engine control switch to RUN. All of the indicating lights except GEN ON on the generator 1 engine/generator control panel should come ON. Check that the generator voltmeter for generator 1 reads 600 volts.
5. Set the frequency switch on the synchronizing control to GEN 1. Turn the frequency trim knob until the frequency meter reads 60 Hertz.
6. Set the generator 1 engine control switch to LOAD.
7. Charge the generator breaker and press the breaker close switch to bring the generator on line to the AC thru bus. The GEN ON light should come on.
a. Close the transformer and auxiliary feeder circuit breakers. Start the essential motors (i.e., air compressors, fuel pumps, radiator fans, etc.) from the motor center.
9. Start the generator 2 engine according to the instructions in steps 1 through 6 above. Check the generator voltmeter for generator 2 and make sure it reads 600 volts.
10. You will now use the synchroscope to synchronize the output frequency of the two generators before you bring generator 2 on line. The synchroscope indicates whether the output frequency of the generator selected by the synchronizing switch is higher or lower than the output frequency of the generator already on line. Set the synchronizing switch on the synchronizing panel to GEN 2. Adjust the frequency trim knob on the generator 2 engine/generator control panel until the synchroscope needle is rotating slowly in-a clockwise direction (about one revolution every 15 seconds).
11. Charge the main generator breaker for generator 2.
12. Watch the synchroscope and press the breaker close switch just before the rotating needle reaches the twelve o'clock position, This brings generator 2 on line to the AC bus. The GEN ON light should come on.
13. The meters on the generator 2 engine/generator control panel will now indicate Generator 2 voltage, current, and power. The values on these meters should be the same for all other on-line generators.
14. Repeat steps 9 through 13 for the additional engine/generator sets.
15. Check that the AC circuit breaker on the front of each power control panel is closed to provide AC power to the SCRs and control logic. At! this point the SCR OK light should be ON and, the SCRs ready to be selected.
16. Energize the appropriate auxiliary motor starters from the motor control center. These would include drill motor blowers, mud pump oiler motors, and so forth. These motors will start automatically when the drilling motor assignments are made from the driller's console. *Drilling* ***motor cut-out switches must be*** *un-pinned* ***before the drive will operate.***
17. The bus A / Bus B tie breaker may be closed using the tie breaker synchronizing indicators.

The rig drive system is now ready for operation.

6-1 .1. DRILLING OPERATION

Once the rig drive system has been started, the drilling motors are controlled from the driller's console and mud pump console. The operator will seldom need to concern himself with the control equipment.

Most operations can be broken down into four types of activities:

- Making motor selections (including alternate assignments)
- Controlling motor speed and direction
- Monitoring rig operation
- Correcting operating problems

These activities are described below. As we describe each activity, refer back to the layout of the driller's console Chapter 3.

6-1.2. MAKING MOTOR SELECTIONS

The selection switches on the driller's console are used to enable SCR power converters for various drilling functions and directions. Each selection switch can select a specific drilling function; for example, mud pump #1, the drawworks, or the rotary table.

Once the motor has been selected, the DRIVE READY light for that motor will come on and the motor can be controlled with the throttle. If the rig has a purge option, the DRIVE READY light will flash until the motor is purged. If the light does not stop flashing, one or more of the following problems exist:

- The motor blower may have failed to start or the signal from it may have failed to reach the power conversion panel.
- The throttle at the driller's console may not be in the OFF position.
- The field or reversing contactor may have failed.
- An auxiliary drive motor (optional) may have failed to start.

CAUTION

If the DRIVE READY light begins to flash during drilling, a problem has arisen. Continued operation under these conditions may cause damage to equipment.

Troubleshooting of this equipment is covered in the μdrill™ 3000 Troubleshooting Guide.

6-1.3. CONTROLLING MOTOR SPEED AND DIRECTION

The driller's console is the main control center for all rig drilling motors. The driller can control both motor speed and direction with the motor throttles and selection switches.

6-1.3.1. Mud Pump. The driller selects a mud pump by turning the proper selection switch to the ON position. The pump is then controlled using the corresponding throttle. The driller can transfer control of the mud pump to that console by turning the selection switch to the REM position. With the selection switch in the REM position, the DRIVE READY lights on the mud pump console will light. If these lights begin flashing, there is a problem (as described above under "Making Motor Selections") that should be corrected. If the lights illuminate continuously, the mud pumps are ready to run.

6-1.3.2. Drawworks. The drawworks are controlled from the driller's console only. The driller controls the drawworks using a single selection switch and a single throttle. The selection switch enables operation of the motor in either the forward (FWD) or reverse (REV) direction. The FWD position of the switch enables regenerative braking of the drawworks motors. Regenerative braking is a method of slowing down a motor by pumping its energy back into the AC power system, thus allowing the motor to decelerate more rapidly.

Drawworks motor speed is controlled by either a hand or foot throttle at the driller's console. Usually, cathead speed is set with the hand throttle and hoist speed is controlled with the foot throttle. Deselection of the drawworks motor must be done in the power control room.

6-1.3.3. Rotary Table. The rotary table is driven by its own motor, which is controlled by a selection switch and throttle on the driller's console. The rotary table torque limit control is used to limit the torque applied to the drill string. This torque limit also limits cathead torque. The μdrill ™ 3000 drive system is the first rig system to allow the driller to control cathead torque.

The rotary table may be chain-driven and clutched off the drawworks motors. In this case, the driller uses the drawwork selection switch and throttle to control the rotary table. Clutching determines the correct current limit value for drawworks or rotary table operation. Whenever the drawworks is unclutched, the rotary table torque limit controls cathead and rotary torque.

6-1.4. MONITORING RIG OPERATION

There are several indicators and meters on the consoles that should be monitored during rig operation. Make a habit of occasionally scanning the console indicators and meters to ensure proper rig operation and early detection of problems.

Should any of the status lights show an abnormal condition the audible alarm will sound, unless the alarm silence switch is ON. Leave the silence switch OFF unless you need to silence an active alarm while you fix a problem.

Before starting a drilling operation or making a function change, the driller should check the following things:

1. Are the correct number of GEN ON lights ON?
2. Are all selection switches set properly and are the correct DRIVE READY lights on?
3. Are all throttles in the OFF position?
4. Is the rotary table torque limit control set according to the drill pipe grade and diameter?
5. Is the drawworks selection switch set in the correct direction (FOR or REV)?

6-1.5. CORRECTING OPERATING PROBLEMS

Two common operating problems are described below along with suggestions for correcting them:

6-1.5.1. Insuff icient Rig Power. If the speed of all DC motors decreases and the KVA/KW limit light comes on, there are not enough engines/generators on line to support the heavy drilling load. Should this occur, the load must be reduced or an additional generator brought on line.

6-1.5.2. Insufficient RT Torque. If only the rotary table's speed decreases, the torque limit set by the rotary table torque limit control has been reached. Ensure that the limit control is set properly for the type of pipe being used. If the setting is correct, you must reduce the torque requirements. There may be too much weight on the bit or the wrong gear may have been selected.

6-1.6. ANCHOR WINDLESS OPERATOR

The anchor windlass motors are individually assigned at the engine room console and each is controlled from an anchor windlass operator console.

When the AW is selected at the engine room the AW console "Drive Ready" light will turn on and the operator will select wildcat or Drive mode. The operator can then select (maintenance) "Power Payout" or "Normal" mode of operation. In Normal mode, the AW can be started with the Jog pushbutton or selected to run in a haul-in or payout direction. The operator can control AW speed and current in either direction.

The AW is stopped by selecting OFF or the Haul-in / OFF / Payout selection switch.

6-l .7. PROPULSION OPERATION

The propulsion motors are assigned at the Engine room console and controlled from either the Engine Room or Pilot House, control transfer is described on the next page.

The propulsion drive ready lights will turn on at the station in control. The operator can control the speed and direction of the propulsion motor with the propulsion throttles.

6-1.7.1. RIG SHUT DOWN. This procedure describes how to completely shut down the rig drive system in preparation for power off maintenance or other shutdown periods.

1. Turn all throttles on the operators consoles to the OFF position. Turn all selection switches to the OFF position.
2. Stop all non-essential loads.
3. Turn the engine control switch on the engine/generator control panel to the RUN position. This unloads the engine. Turn the generator circuit breaker to the OFF position. Repeat this step for each generator.
4. Turn the engine control switch on each engine/generator control panel to the IDLE position. After the manufacturer's recommended idle time, turn the engine control switch to the OFF position.
5. Turn the battery module power switch to the OFF position.

6-2. CONTROL TRANSFER

6-2.1. ENGINE ROOM TO PILOT HOUSE

Initially, the "Control Here" light will be on at the engine room and the "Engine Room Control" light will be on at the pilot house.

The engine room operator will request a control transfer by turning his control location selector switch to the "Pilot House" position. The engine room remains in control.

The "take control" light will turn on at the Pilot House. After both the Engine Room and Pilot House throttles are zeroed, the pilot house operator will push his "Take Control" pushbutton. Control will transfer to the pilot house.

The Engine room control here light will turn OFF and the "Control Pilot House" light will turn on.

The Pilot House "Engine Room Control" light will turn off and the "Pilot House Control" light will turn on. The pilot house propulsion throttles are now active.

6-2.2. PILOT HOUSE TO ENGINE ROOM TRANSFER

The Pilot House and engine room operators will zero the propulsion throttles. The Engine room operator will then turn the control location switch from "Pilot House" to "Here". Control will immediately transfer.

The Pilot House "Engine Room Control" light will turn ON and the Engine Room "CONTROL HERE" light will turn on.

The Engine Room propulsion throttles are now active.



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